Slater's Rules allow you to calculate the **effective nuclear charge (Z_{eff})**, which is the net positive charge experienced by a specific electron in a multi-electron atom. In **Photoelectron Spectroscopy (PES)**, the positions of the peaks represent the **binding energy (BE)** of the electrons in different subshells. [1, 2, 3]

Because an electron's binding energy depends directly on how strongly it is attracted to the nucleus, Z_{eff} provides the theoretical bridge to estimate and understand these PES peak positions. [1, 3]

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1. Calculating Z_{eff} using Slater's Rules

The effective nuclear charge is calculated using the formula:

$$Z_{eff} = Z - S$$

Where Z is the atomic number (actual number of protons) and S is the shielding (or screening) constant. [2, 4]

Step-by-Step Rules for S:

1. **Group the electron configuration** in the following specific order:
   $$(1s) \ (2s, 2p) \ (3s, 3p) \ (3d) \ (4s, 4p) \ (4d) \ (4f) \ (5s, 5p) \dots$$
2. **Select the electron of interest.** Electrons in groups to its right (higher energy) provide **0 shielding**. [5]
3. **If the electron of interest is in an ns or np orbital:**
   - Other electrons in the **same group** shield by **0.35** each (except for $1s$, where the other electron shields by **0.30**).
   - Electrons in the $(n-1)$ **shell** shield by **0.85** each.
   - Electrons in the $(n-2)$ **or lower shells** shield completely by **1.00** each. [2, 6]
4. **If the electron of interest is in an nd or nf orbital:**
   - Other electrons in the **same group** shield by **0.35** each.
   - **All** electrons in groups to the left (inner shells and lower-energy subshells) shield by **1.00** each. [2, 5]

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2. Linking Z_{eff} to Binding Energy (BE) on a PES

According to a simplified version of Koopmans' theorem, the binding energy required to eject a specific electron can be approximated using a modified Bohr/hydrogen-like orbital energy equation: [7, 8]

$$BE \approx 13.6eV \times \frac{Z_{eff}^2}{n^2}$$

- **Higher Z_{eff} = Stronger nuclear pull = Higher Binding Energy**. [1, 3]
- On a PES spectrum, peaks with the highest binding energy (found closest to the nucleus, e.g., $1s$) have a very low shielding constant S, a massive Z_{eff}, and a small principal quantum number n. [1, 9]
- Conversely, valence electrons are highly shielded by core electrons (high S), resulting in a low Z_{eff} and a **low binding energy** peak. [1, 9]

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3. Concrete Example: Carbon ($Z = 6$)

Let's see how Slater's rules map to the three distinct peaks observed on a PES spectrum for Carbon ($1s^2 2s^2 2p^2$).

Subshell (PES Peak)	Slater Grouping & Math	Z_{eff} Calculation	Estimated BE Formula Component	PES Location
$1s$ (Core)	$(1s^1) \dots$ One other electron in the same group. $S = 1 \times 0.30 =$	$Z_{eff} = 6 - 0.30 =$	$\propto \frac{5.70^2}{1^2} = 32.49$	**Highest BE peak** (Far left)
$2s/2p$ (Valence)*	$(1s^2)(2s^2 2p^1) \dots$ 3 other valence electrons, 2 core electrons. $S = (3 \times 0.35) +$	$Z_{eff} = 6 - 2.75 =$	$\propto \frac{3.25^2}{2^2} = 2.64$	**Lowest BE peaks** (Far right)

**Note: Slater's rules group $2s$ and $2p$ together, giving them the same Z_{eff}. In reality, $2s$ penetrates closer to the nucleus than $2p$, meaning $2s$ experiences slightly less shielding, a slightly higher true Z_{eff}, and appears as a separate peak slightly to the left of $2p$ on a real PES graph.* [2]

[1] https://www.khanacademy.org

[2] https://chem.libretexts.org
[3] https://www.pearson.com
[4] https://www.youtube.com
[5] https://www.youtube.com
[6] https://en.wikipedia.org
[7] https://pubs.aip.org
[8] https://www.nat.vu.nl
[9] https://www.youtube.com